LAB SHEET 7

Name:
USN: Date:

PRECISION RECTIFIER

Aim: To design, build and test Precision Rectifiers for given specifications.

1. To understand the concept of super diode.
2. To implement inverting and non-inverting half wave rectifier.
3. To implement inverting and non-inverting full wave rectifier.
4. Plot input and output waveforms.

Components/Parts required: µA741 IC (Op-Amp), resistors, diodes (IN4007), signal generator, output meter, DC power supply.

Prelab:

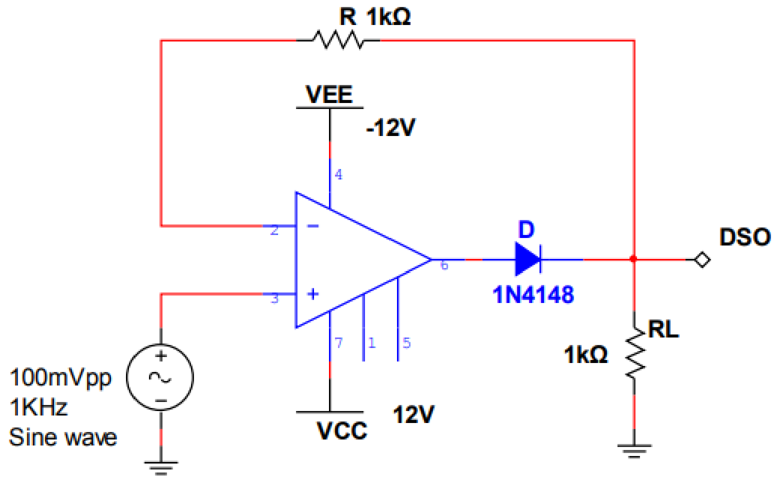
Theory:

A rectifier is a DC voltage source. It has finite output impedance in series with a DC voltage source. Therefore a rectifier can be modelled as a constant voltage source in series with a resistance. The rectification becomes a crucial problem when we have to rectify millivolt signal at high frequencies. This is because the conventional diodes require at least 0.2 V (Ge) or 0.7 V (Si) across them for a forward bias and hence a millivolt signal cannot forward bias a conventional diode. Also the switching speed of conventional diode is very low. If we use a diode in the forward path of op amp cut in voltage of the diode will be divided by the large open loop gain of op amp, thus making the diode as ideal one i.e. a diode with zero cut in voltage. When we use a conventional diode in the feedback path of an op amp, we can rectify millivolt signals at high frequencies. This set up is then called as Precision rectifiers. The combination of diode and op-amp in precision rectifier circuit is called as Superdiode. In case of conventional rectifier, output peak will be less than the input peak. This is due to loss of voltage across the diode due to finite cut in voltage. But in the case of precision rectifier diodes, it will have zero cut in voltage and hence output peak will be same as input peak. Precision rectifiers are classified as Half wave precision rectifier and Full wave precision rectifier according to the operation of the circuit.

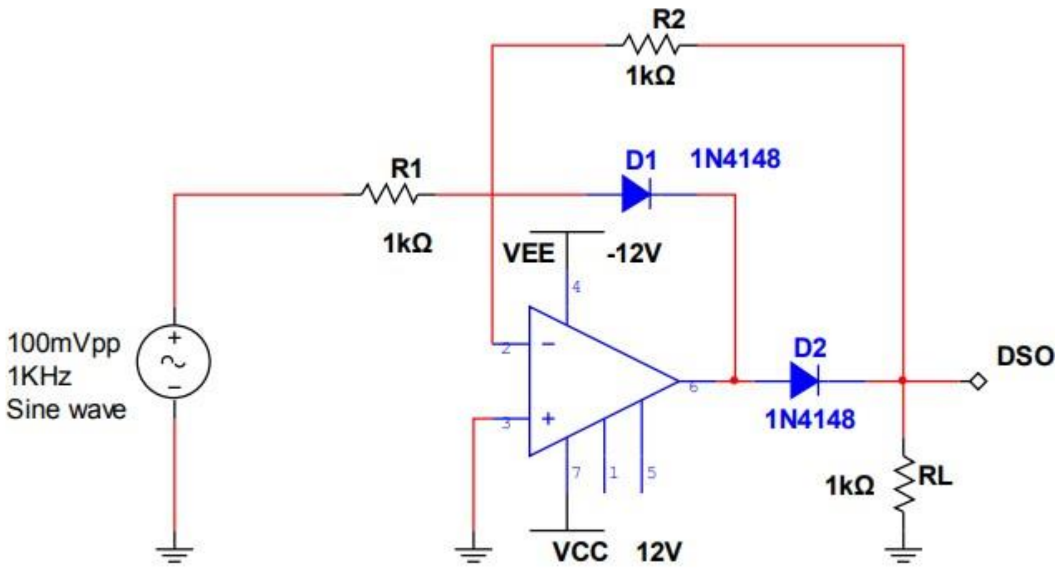
Procedure:

1) Connect the circuit as per the circuit diagram
2) Give a sinusoidal input of 100mVPP, 1 kHz from a signal generator
3) Switch on the power supply and note down the output from CRO
4) Repeat the above procedure by reversing the diodes
5) Measure the output voltage for positive and negative half cycle
6) Draw input and output waveforms for each circuit

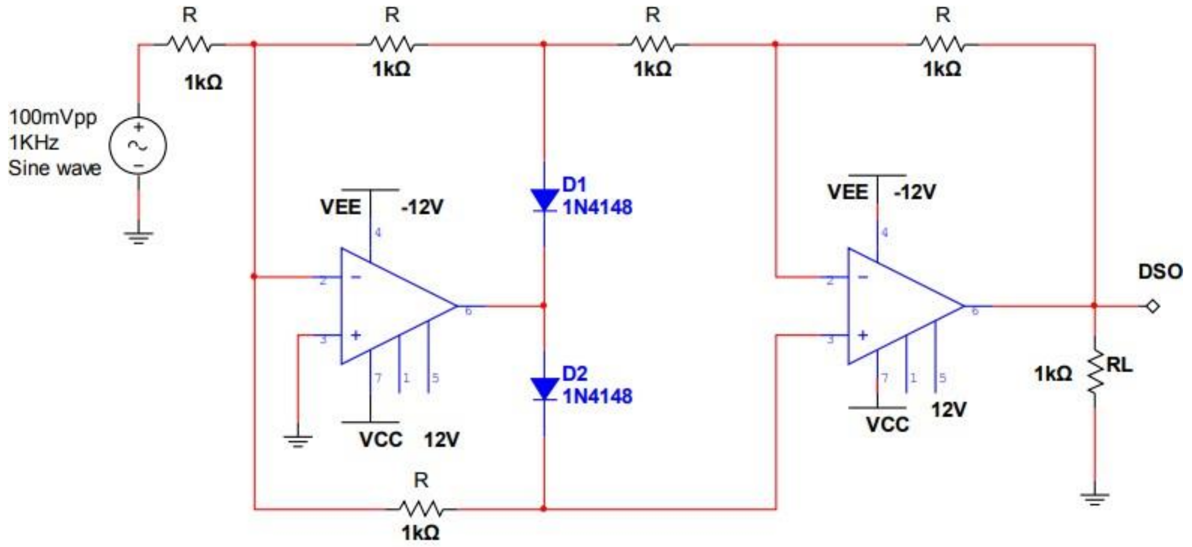
Circuit Diagram:-



(1) Non Inverting Half Wave Precision Rectifier



(2) Inverting Half Wave Precision Rectifier



(3) Full Wave Precision Rectifier

Design Question:

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Observation Table

$V_{in} = 100\ mV_{pp}$

Rectifier Type	Output voltage (V_o)	
	For +ve half Cycle	For - ve half Cycle
Non Inverting Half Wave Precision Rectifier		
Inverting Half Wave Precision Rectifier		
Full Wave Precision Rectifier		

***Draw the expected input and output waveforms, transfer characteristics**

RESULTS:

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Simulation (LTSPice)

Sl. No	Desired Gain (V/V)	Resistor values (theoretical)
1	1	
2	1.5 (positive cycle), 1 (negative cycle)	
3	2 (both cycles)	
4	5 (Both cycles)	

Hardware

$V_{in} = 100\ mV_{pp}$

Rectifier Type	Output voltage (V_o)	
	For +ve half Cycle	For - ve half Cycle
Non Inverting Half Wave Precision Rectifier		
Inverting Half Wave Precision Rectifier		
Full Wave Precision Rectifier		

***Attach the simulation results and virtual lab report with the lab sheet**
***Attach the graph sheets with input and output waveforms**

Inference: